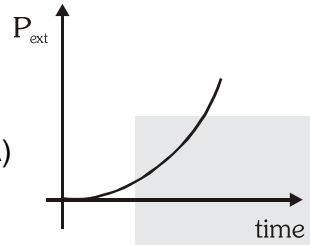
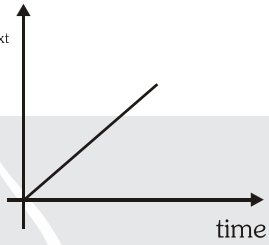
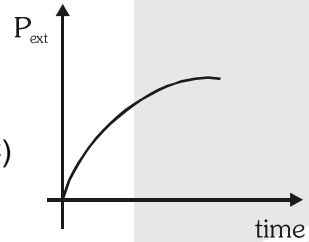
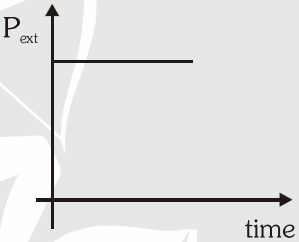
SYLLABUS : ROTATIONAL BODY DYNAMICS

1. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :–

(A) 

(B) 

(C) 

(D) 

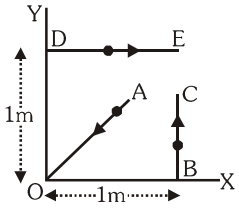
2. A body is rolling without slipping on a horizontal surface and its rotational kinetic energy is equal to the translational kinetic energy. The body is :–

(A) disc (B) sphere (C) cylinder (D) ring

3. A uniform metre stick is held vertically with one end on the floor and is allowed to fall. The speed of the other end when it hits the floor assuming that the end at the floor does not slip :

(A) $\sqrt{4g}$ (B) $\sqrt{3g}$ (C) $\sqrt{5g}$ (D) $\sqrt{g}$

4. A particle of mass m moves with a constant velocity. Which of the following statements is not correct about its angular momentum about point O:



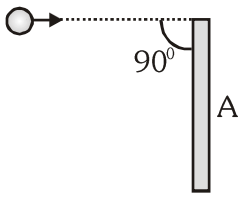
(A) it is zero when it is at A and moving along OA

(B) the same at all points along the line DE

(C) of the same magnitude but oppositely directed at B and D

(D) increases as it moves along the line BC

5. A thin rod of mass M and length L is struck at one end by a ball of clay of mass m, moving with speed v as shown in figure. The ball sticks to the rod. After the collision, the angular momentum of the clay-rod system about A, the midpoint of the rod, is



(A) $\left(m+\frac{M}{3}\right)\left(\frac{vL}{2}\right)$ (B) $\left(m+\frac{M}{12}\right)\left(\frac{vL}{2}\right)$ (C) $\frac{mvL}{2}$ (D) mvL

6. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 sec. the magnitude of this torque is :

(A) $4A_0$ (B) A_0 (C) $3A_0/4$ (D) $12A_0$

7. A particle is projected at time t = 0 from a point P on the ground with a speed V_0, at an angle of 45° to the horizontal. What is the magnitude of the angular momentum of the particle about P at time t= v_0/g.

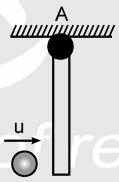
(A) $\frac{mv_0^2}{2\sqrt{2}g}$ (B) $\frac{mv_0^3}{\sqrt{2}g}$ (C) $\frac{mv_0^2}{\sqrt{2}g}$ (D) $\frac{mv_0^3}{2\sqrt{2}g}$

8. A boy sitting firmly over a rotating stool has his arms folded. If he stretches his arms, his angular momentum about the axis of rotation

(A) increases (B) decreases

(C) remains unchanged (D) doubles

9. In the given figure a ball strikes a rod elastically and rod is smoothly hinged at point A. Then which of the statement(s) is/are correct for the collision?



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(A) linear momentum of system (ball + rod) is conserved

(B) angular momentum of system about hinged point A is conserved

(C) initial KE of the system is equal to final KE of the system

(D) linear momentum of ball is conserved.

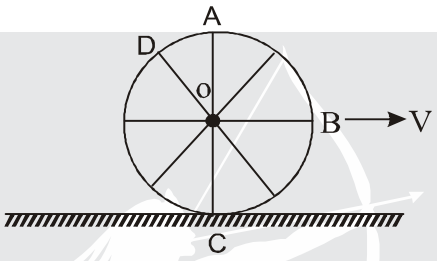
10. A cord is wound over a cylinder of radius r and moment of inertia I. A mass m is attached to the free end of the cord. The cylinder is free to rotate about its own horizontal axis. If mass m is released from rest, then the velocity of the mass after it had fallen through a distance h will be-

(A) $(2gh)^{1/2}$ (B) $\left(\frac{2mghr^2}{I}\right)^{1/2}$ (C) $\left(\frac{2mghr^2}{I+mr^2}\right)^{1/2}$ (D) $\left(\frac{mghr^2}{I+2mr^2}\right)^{1/2}$

11. The centre of a disc rolling without slipping on a plane surface moves with speed u. A particle, on the lower half of the rim making an angle 60º with vertical, will be moving at speed

(A) zero (B) u (C) $\sqrt{2}u$ (D) 2u

12. Consider a disc rolling without slipping on a horizontal surface at a linear speed V as shown in figure



(A) the speed of the particle A is 2V

(B) the speed of B, C and D are all equal to V

(C) the speed of C is zero and speed of B is $\sqrt{2}V$

(D) the speed of O is less than the speed of B

13. When a bicycle is in motion (accelerating) on a rough horizantal plane, the force of friction exerted by the plane on the two wheels is such that it acts :

(A) In the backward direction on the front wheel and in the forward direction on the rear wheel, when the speed increases.

(B) In the forward direction on the front wheel and in the backward direction on the rear wheel, when brakes are applied on rear wheel.

(C) In the backward direction on both front and the rear wheels, when brakes are applied on both wheels.

(D) In the forward direction on both the front and the rear wheels

14. A hollow sphere and a solid sphere having equal mass and equal radii are rolled down without slipping on a rough inclined plane.

(A) The two spheres reach the bottom simultaneously

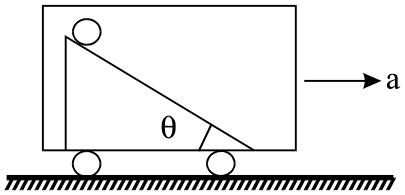
(B) The hollow sphere reaches the bottom with lesser speed.

(C) The solid sphere reaches the bottom with greater kinetic energy

(D) The two spheres will reach the bottom with same linear momentum

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15. A smooth inclined plane fixed in a car accelerating on a horizontal road is shown in figure. The angle of incline θ is related to the acceleration a of the car as $a = g \tan\theta$. If a rigids sphere is set in pure rolling on the incline



(A) it will continue pure rolling (B) Friction will act on it

(C) its angular velocity will increase (D) its angular velocity will decrease.

16. A body is given translational velocity and kept on a surface that has sufficient friction. Then:

(A) body will move forward before pure rolling

(B) body will move backward before pure rolling

(C) body will start pure rolling immediately

(D) none of these

17. If the earth were to suddenly contract to $\frac{1}{n}$th of its present radius without any change in its mass then the duration of the new day will be nearly :–

(A) $\frac{24}{n}$ hour (B) 24n hour (C) $\frac{24}{n^2}$ hour (D) $24n^2$ hour

18. A thin circular ring of mass M and radius 'r' is rotating about its axis with a constant angular velocity ω. Four objects each of mass m, are kept gently to the opposite ends of two perpendicular diameters of the ring. The new angular velocity of the ring will be :–

(A) $\frac{M\omega}{4m}$ (B) $\frac{M\omega}{M+4m}$ (C) $\frac{(M+4m)\omega}{M}$ (D) $\frac{(M+4m)\omega}{M+4m}$

19. A sphere is released on a smooth inclined plane from the top. When it moves down its angular momentum is:

(A) conserved about every point

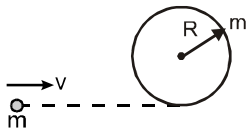
(B) conserved about the point of contact only

(C) conserved about the centre of the sphere only

(D)conserved about any point on a fixed line parallel to the inclined plane and passing through the centre of the ball.

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20. A circular wooden loop of mass m and radius R rests flat on a horizontal frictionless surface. A bullet, also of mass m, and moving with a velocity V, strikes the loop and gets embedded in it. The thickness of the loop is much smaller than R. The angular velocity with which the system rotates just after the bullet strikes the loop is

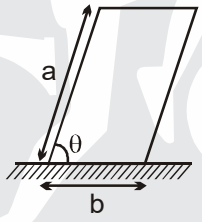


(A) $\frac{V}{4R}$ (B) $\frac{V}{3R}$ (C) $\frac{2V}{3R}$ (D) $\frac{3V}{4R}$

21. A uniform cube of side a and mass m rests on a rough horizontal table. A horizontal force 'F' is applied normal to one of the faces at a point that is directly above the centre of the face, at a height $\frac{3a}{4}$ above the base. The minimum value of 'F' for which the cube begins to tilt about the edge is (assume that the cube does not slide).

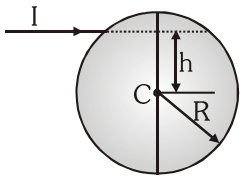
(A) $\frac{2}{3}mg$ (B) $\frac{4}{3}mg$ (C) $\frac{5}{4}mg$ (D) $\frac{1}{2}mg$

22. A homogenous block having its cross-section to be a parallelogram of sides 'a' and 'b' (as shown) is lying at rest and is in equilibrium on a smooth horizontal surface. Then for acute angle θ :



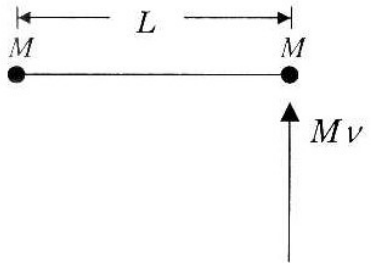
(A) $\cos\theta \le \frac{b}{a}$ (B) $\cos\theta \ge \frac{b}{a}$ (C) $\cos\theta < \frac{b}{a}$ (D) $\cos\theta > \frac{b}{a}$

23. A solid sphere is placed on a horizontal plane. A horizontal impulse I is applied at a distance h above the central line as shown in the figure. Soon after giving the impulse the sphere starts rolling. The ratio h/R would be-



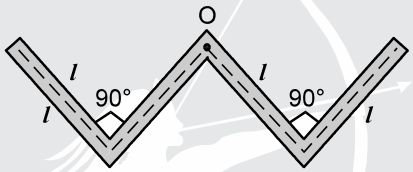
(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$

24. Two particles, each of mass M, are connected by a rod of negligible mass and length L. The system is lying on a horizontal frictionless surface. An impulse Mv, perpendicular to the rod, is given at one end of the rod as shown in the figure. The angular velocity acquired by the rod is



(A) $\frac{4v}{L}$ (B) $\frac{2v}{L}$ (C) $\frac{v}{L}$ (D) $\frac{v}{4L}$

25. A thin uniform rod of length 4 *l*, mass 4m is bent at the points as shown in the fig. What is the moment of inertia of the rod about the axis passing point O & perpendicular to the plane of the paper.



(A) $\frac{m\ell^2}{3}$ (B) $\frac{10m\ell^2}{3}$ (C) $\frac{m\ell^2}{12}$ (D) $\frac{m\ell^2}{24}$

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ANSWER KEY

1.	(B)	**2.**	(D)	**3.**	(B)	**4.**	(D)	**5.**	(C)
6.	(C)	**7.**	(D)	**8.**	(C)	**9.**	(BC)	**10.**	(C)
11.	(B)	**12.**	(ACD)	**13.**	(ABC)	**14.**	(B)	**15.**	(A)
16.	(A)	**17.**	(C)	**18.**	(B)	**19.**	(D)	**20.**	(B)
21.	(A)	**22.**	(A)	**23.**	(B)	**24.**	(C)	**25.**	(B)